What to measure on the forklift (ME-01, ME-02)

Every size of the forklift in the code is a **guess** right now. Only the box size of the Huina 1577 Pro (524 × 187 × 317 mm) is real. The same numbers are used in four places:

Where	What uses the numbers
`ai/training/forklift_rl/config.yaml`	the docking AI is trained on a forklift with these sizes
`robot/cad/params.yaml`	the 3D-printed mounts are drawn to fit them
`robot/ros2_ws/.../dimensions.yaml`, `nav2_params.yaml`	the robot model, the footprint, the turning radius
`robot/firmware/.../config.h`	wheelbase, steering limit, lift range, speed

You measure; I put each number in all the right files, retrain the docking AI (about 15 minutes) and re-draw the mounts. **You don't need to edit any code.**

Before you start

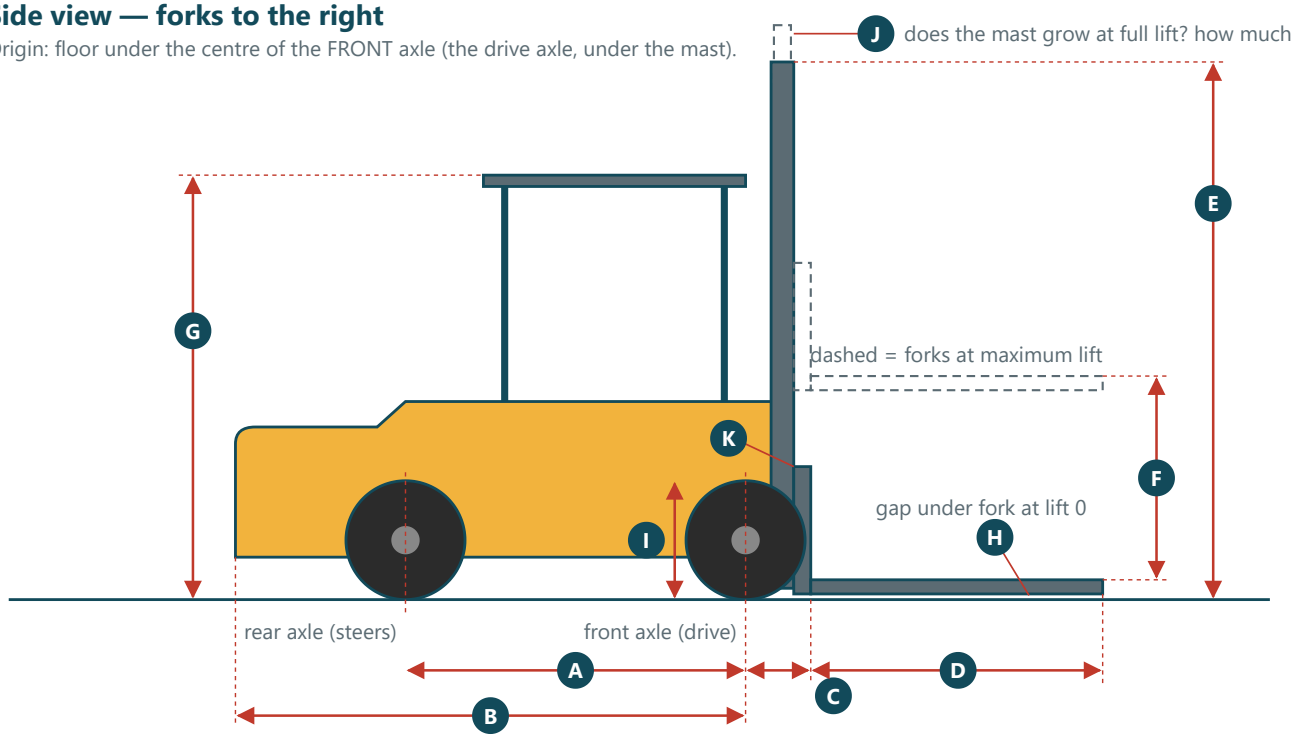
- **Tools:** digital calipers, a tape measure, a phone protractor app, a stopwatch, a multimeter (or the INA219), a kitchen scale, a pen and some tape.
- **Units:** millimetres with one decimal for small parts (calipers), whole millimetres for long ones (tape).
- **Photos:** take a photo of every place you measure, with the calipers or tape visible. If a number looks strange later, the photo answers the question without opening the forklift again.
- **If a part is shaped differently from the drawings**, measure what makes sense and send a photo. The drawings are schematic; they show *which* distance, not the exact shape.
- **The drive axle is at the front (under the mast) and the steering is at the back** in the drawings. If your forklift is the other way round, say so — that changes the model, not only the numbers.

Most urgent: group 1. With only those numbers I can already retrain the docking AI.

Group 1 — Body and forks (needed to retrain the AI)

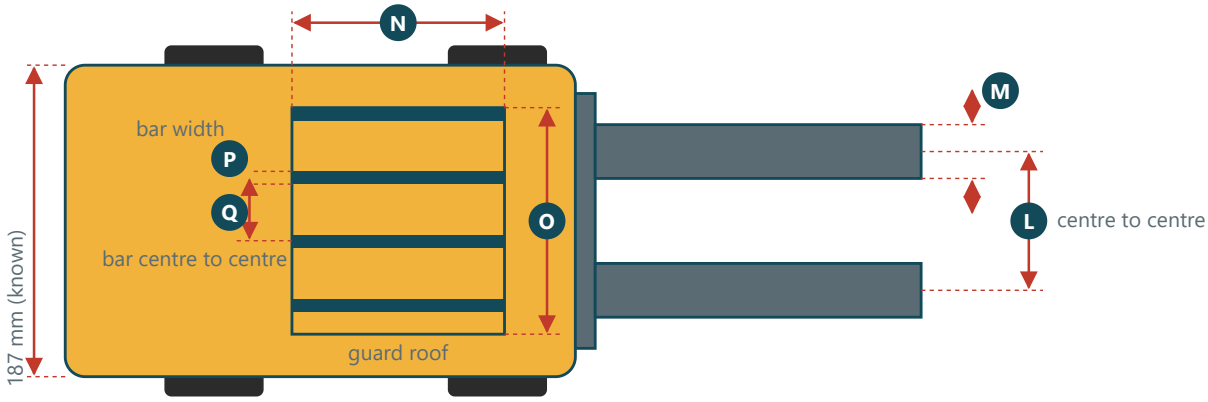
Side view — forks to the right

Origin: floor under the centre of the FRONT axle (the drive axle, under the mast).



Total length 524 mm, width 187 mm and height 317 mm are already known (Huina 1577 Pro box).

Top view — forks to the right



Front view of the forks (looking at the fork tips)



Calipers for L, M, R, P (mm, 1 decimal).
Measure L between the CENTRES of the forks: outer edge to outer edge minus one fork width.

ID	What	How	Current guess
A	Wheelbase: front axle centre → rear axle centre	tape, between the wheel hubs, on both sides; average	200 mm

ID	What	How	Current guess
B	Body behind the front axle: front axle centre → very back of the body	tape	300 mm
C	Front axle centre → front face of the fork carriage (where the forks hang)	tape or calipers	30 mm
D	Fork length: carriage face → fork tip (the flat part only)	calipers / tape	180 mm
E	Mast height: floor → top of the mast, forks down	tape	317 mm
F	Lift range: fork top at lowest → fork top at highest	lift fully, measure both heights from the floor, subtract	120 mm
G	Floor → top of the overhead guard roof	tape	245 mm
H	Gap between the floor and the fork bottom, forks fully down	calipers (depth rod) or feeler / stack of paper	4 mm
I	Wheel diameter (front, driven wheel)	calipers across the tyre	70 mm
J	Does the inner mast rise above E at full lift? By how much? (0 if not)	tape at full lift	0 mm
K	Backrest height: fork top → top of the carriage backrest	calipers	70 mm
L	Fork spacing, **centre to centre**	calipers: outer edge to outer edge, minus one fork width	90 mm
M	Fork width (one fork)	calipers	35 mm
R	Fork thickness	calipers	12 mm

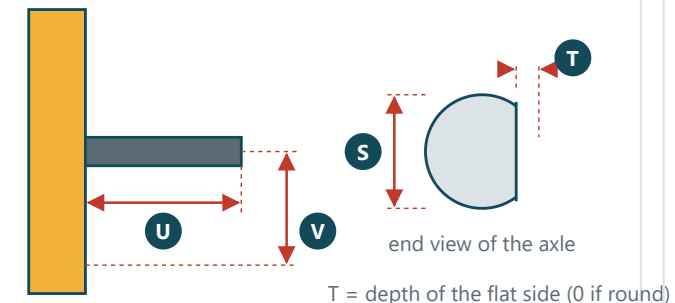
Group 2 — Steering, speed and motors (the controller and the AI need these)

ID	What	How	Current guess
SA	Maximum steering angle, left and right	full lock; phone protractor on the rear wheel against the body side	30°
TC	Turning circle	see panel 4 below: full lock, drive slowly one full circle with a pen taped under the **front-axle centre**, measure the circle's diameter. Do it left and right	≈ 700 mm
SP	Top speed	full throttle on the stock remote over 2 m of floor, time it 3 times: speed = 2 / time	0.45 m/s
GR	Gear ratio	mark the wheel and the axle stub; turn the wheel exactly one turn by hand; count axle-stub turns	1
LS	Lift speed	time the forks from lowest to highest, 3 times	≈ 2.4 s
ME-01	Which motors are inside	for each — drive, steering, lift, side-shift: is it a DC motor or a servo? how many wires? any label/number?	—
ME-02	Motor currents	multimeter in series (or INA219) at 7.4 V: current while **running** free, and **stall** current (hold the wheel/fork for **under 2 seconds**). Drive, steering, lift	—

ID	What	How	Current guess
WT	Weight of the bare forklift	kitchen scale	—

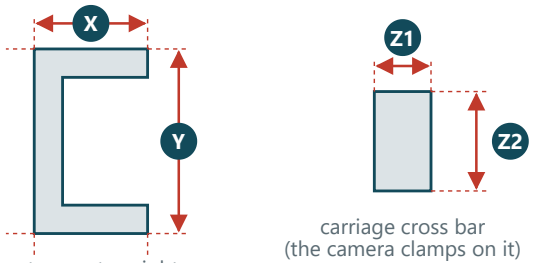
Group 3 — Where the printed mounts attach

1 · Drive axle end (encoder magnet goes here)



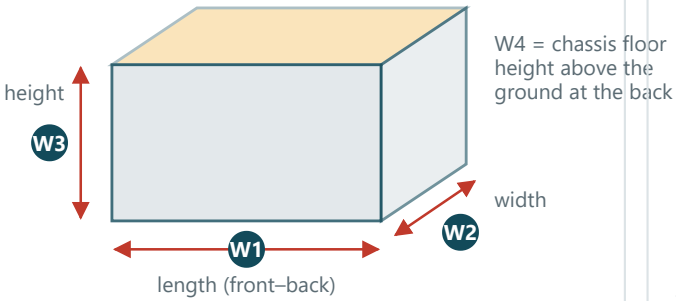
T = depth of the flat side (0 if round)

2 · Mast rail and carriage bar (cut through)



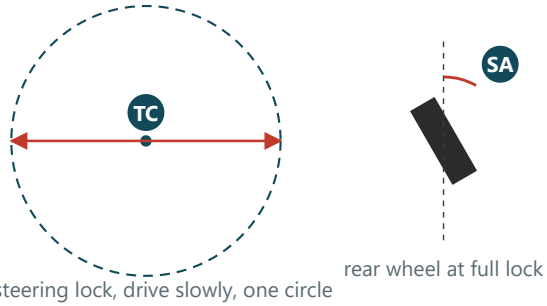
X = depth (front to back), Y = width (side to side)

3 · Free space under the rear (yellow) cover



4 · Steering angle and turning circle (top view)

TC = diameter of the circle the front-axle centre draws

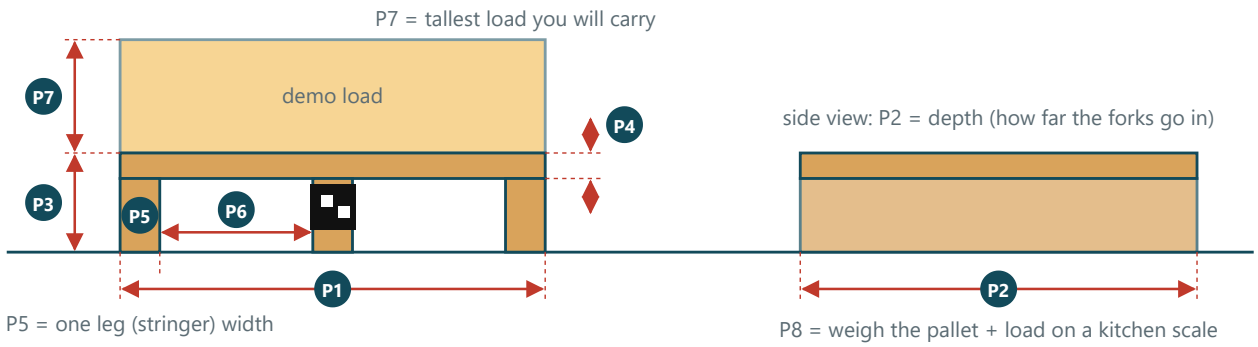


ID	What	How	Current guess
S	Drive axle stub diameter (the encoder magnet cap presses on it)	calipers	4.0 mm
T	Depth of the flat side on the axle (0 if it is round)	calipers: full diameter minus the across-the-flat size	0.5 mm
U	Axle end → chassis face it sticks out of	calipers, depth rod	12 mm
V	Axle centre → a flat chassis surface a small bracket could be screwed to	calipers; photo of that surface	14 mm
X, **Y**	Outer mast upright (channel) depth and width — the slide-pot bracket clamps on it	calipers	10 × 14 mm
Z1, **Z2**	Carriage cross bar thickness and height — the camera clamp grips it	calipers	6 × 12 mm
N, **O**	Guard roof length and width (LiDAR sits on it)	calipers	120 × 110 mm

ID	What	How	Current guess
P, Q	Guard roof bar width, and centre-to-centre distance between bars	calipers	8 mm, 30 mm
W1–W3	Free space under the rear (yellow) cover: length × width × height — the Jetson box goes there	open the cover, calipers/tape	150 × 160 × 75 mm
W4	Height of the chassis floor above the ground at the back	calipers	25 mm

Group 4 — Pallet

Pallet — front view (the side the forks enter) and side view



ID	What	Current guess
P1	Width (side the forks enter)	200 mm
P2	Depth (how far the forks go in)	200 mm
P3	Total height (floor → top of the deck)	35 mm
P4	Top deck thickness	8 mm
P5	Width of one leg (stringer)	20 mm
P6	Width of one fork opening (between two legs)	70 mm
P7	Height of the tallest load you will put on a pallet	40 mm
P8	Weight of a pallet with its load	250 g

Measure P6 and the opening **height** (floor → underside of the deck) carefully: the forks plus the fork-tip sensor clip have to fit inside with 2 mm to spare.

Group 5 — The parts you received (check against the datasheet)

When the parts arrive, measure the boards with calipers. Each one has a bracket drawn for it, so the screw holes have to match the real board.

Part	Measure
VL53L1X ToF board (×6)	board W × H, the 2 screw-hole centres
Jetson Orin Nano carrier	distance between the 4 mounting holes, hole → board corner
AS5600 encoder board	board size, hole centres
Slide potentiometer module	body L × W × H, knob travel, hole centres
4-slot 18650 holder + BMS	L × W × H of each
E-Stop contact block	size under the panel
RPLIDAR C1	hole spacing on the bottom, height of the scan window above the base

Group 6 — The warehouse floor (later, EN-01 … EN-08)

- The room: length × width of the floor area, where the walls are.
- Each rack: position of its front-left corner from one fixed corner of the floor, its width and depth, and the height of every shelf level.
- Each charger spot and each zone (inbound, outbound, staging, no-go): position and size.
- Aisle widths in front of the racks (≥ 85 cm needed) and between racks.
- A sketch on squared paper with these numbers and a photo is enough. I turn it into `warehouse_map.yaml` and the Gazebo world.

Sending the numbers back

Copy this block, fill it in, and send it with the photos. Leave a line empty if you couldn't measure it.

```
forklift: Huina 1577 Pro    drive axle at: front / back    steering at: front / back

A  wheelbase                      =     mm
B  body behind front axle         =     mm
C  front axle -> carriage face    =     mm
D  fork length                    =     mm
E  mast height                    =     mm
F  lift range                     =     mm   (lowest    mm, highest    mm)
G  guard roof top                 =     mm
H  fork gap to floor              =     mm
I  wheel diameter                 =     mm
J  mast rise at full lift         =     mm
K  backrest height                =     mm
L  fork spacing centre-centre     =     mm
M  fork width                     =     mm
R  fork thickness                 =     mm

SA steering angle left / right    =     ° /     °
TC turning circle left / right    =     mm /     mm
SP 2 m times                      =     s,     s,     s
GR axle turns per wheel turn      =
LS lift times                     =     s,     s,     s
WT forklift weight                =     g
motors (type, wires, label):  drive:           steering:           lift:           side-shift:
currents running / stall (A): drive:     /     steering:     /     lift:     /

S  axle diameter = mm   T flat = mm   U axle end->face = mm   V axle->flat surface = mm
X x Y mast rail = x mm   Z1 x Z2 carriage bar = x mm
N x O roof = x mm   P bar width = mm   Q bar pitch = mm
W1 x W2 x W3 rear space = x x mm   W4 rear floor = mm

P1 = mm  P2 = mm  P3 = mm  P4 = mm  P5 = mm  P6 = mm  opening height = mm  P7 = mm  P8 = g
```